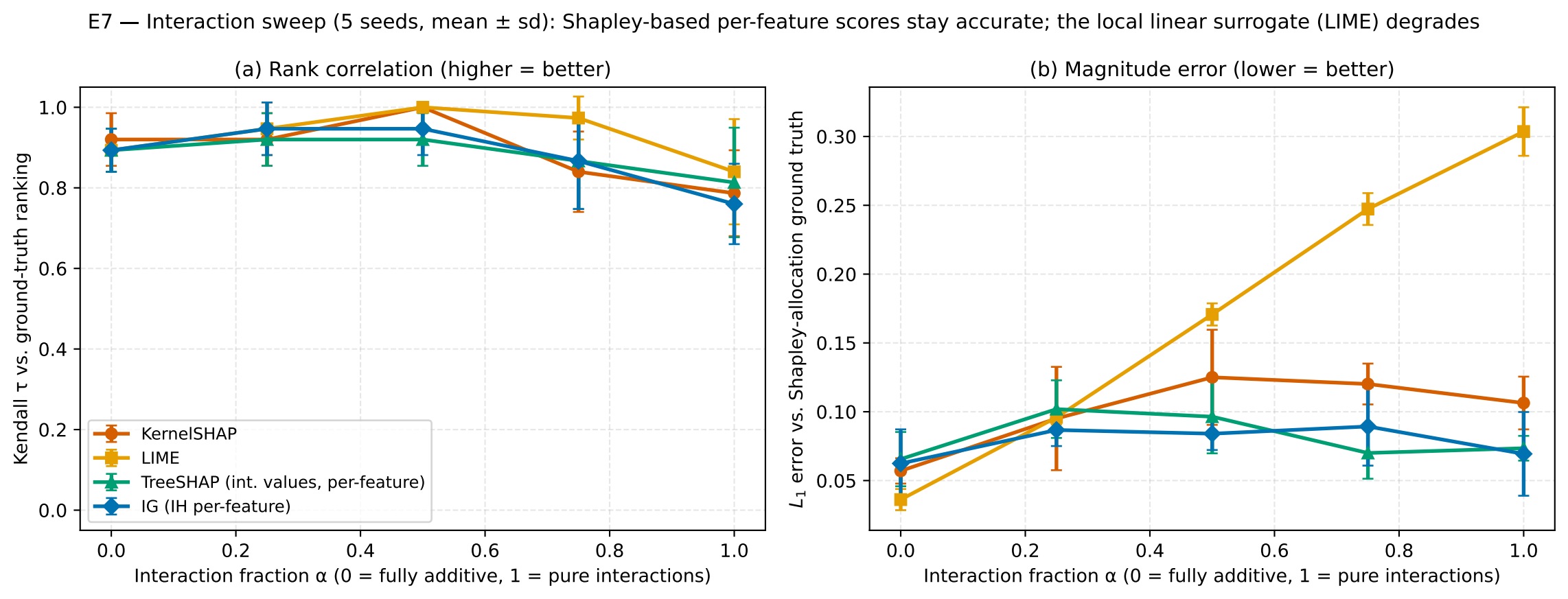
E7 — Interaction sweep (5 seeds, mean ± sd): Shapley-based per-feature scores stay accurate; the local linear surrogate (LIME) degrades
(a) Rank correlation (higher = better)
Kendall τ vs. ground-truth ranking
1.0
0.8
0.6
0.4
0.2
0.0
KernelSHAP
LIME
TreeSHAP (int. values, per-feature)
IG (IH per-feature)
0.0
0.2
0.4
0.6
0.8
1.0
Interaction fraction α (0 = fully additive, 1 = pure interactions)
(b) Magnitude error (lower = better)
L_1 error vs. Shapley-allocation ground truth
0.30
0.25
0.20
0.15
0.10
0.05
0.0
0.2
0.4
0.6
0.8
1.0
Interaction fraction α (0 = fully additive, 1 = pure interactions)